

Supplementary material — Unknown is not normal

S1. Calculator readings and clinical review

Each calculator cites its primary source in code. Where the source was ambiguous or differed from common implementations, the implemented reading and the alternatives are listed below. The author, an emergency physician, reviewed and accepted the implemented readings.

Type:

- **S (source):** resolvable by reading the full text of the primary paper. These cluster on the abstract-only [A] sources, PERC and Wells 2000.
- **C (clinical):** needs clinical judgement.
- **M (modelling):** a modelling choice with no clinical answer.

Priority is face validity: how much a clinician reading a generated case would notice if it were wrong.

- **High:** visible in every case of that calculator.
- **Med:** visible in some cases.
- **Low:** internal only.

For a bounds benchmark most items only need to be *fixed and documented*: the ground truth is self-consistent either way.

#	Type	Priority	Calculator	Item	Implemented	Alternatives / notes
1	S	High	HEART	Troponin bands	$\leq$ normal / $1-3\times$ / $> 3\times$, deliberately following the validation cohorts (Backus 2010/2013, Poldervaart 2013), not the 2008 table	Six 2008: $1-2\times$ / $> 2\times$. Exactly $3\times$ at the band edge: $\geq 3\times$ (Backus 2011, Poldervaart) vs $> 3\times$ (MDCalc)
27	C	High	HEART	Troponin assay reference	Troponin is a level relative to the assay's upper reference limit, accepted by the author. Checked: every HEART note states the level relative to the reference limit; the one raw value also gives the reference limit	Real hs-troponin uses sex-specific 99th percentiles. Out of scope while troponin is a ratio; needed for real data
23	S	High	HEART	Age exactly 65	2 points (age ≥ 65), per the 2008 text ("65 years or older")	2008 table says " ≤ 65 " (typo); some renderings use $45-65 \square 1$, $> 65 \square 2$ (65 scores 1)
2	S	High	HEART	Age exactly 45	1 point (age ≥ 45)	Poldervaart 2013: $\leq 45 \square 0$
4	C	High	HEART	Smoking recency	≤ 3 months (MDCalc), accepted by the author	< 1 month (Six 2008). No note states a quit date, so the choice affects no case

#	Type	Priority	Calculator	Item	Implemented	Alternatives / notes
5	C	High	HEART	Family history	first-degree relative with CAD before 65, accepted by the author	Only 3 notes give an age (mother 54, brother 55, father 58), all clear-cut, so the choice affects no case not defined in Six 2008; BMI > 30 in Poldervaart 2013
6	S	Med	HEART	Obesity	boolean, labelled BMI > 30	Six 2008 lists revascularisation, MI, stroke, PAD. MedCalc-Bench counts TIA as an ordinary risk factor (1 of 100 anchor disagreements)
7	S	Med	HEART	TIA as atherosclerotic disease	included (2 points)	2008 table: ST depression only
3	S	Med	HEART	ECG 2 points	significant ST deviation (depression or elevation)	2004 abstract: rule needs > 94%. "Sats normal on 2 L" is normal but PERC-positive in some readings; our notes never render supplemental O2
11	S	High	PERC	SaO2	positive if < 95%, on room air (the renderer states "on room air")	requiring general anaesthesia (some sources)
12	S	Med	PERC	Surgery/trauma	within 4 weeks requiring hospitalisation (2008 abstract)	any hormone therapy (2004 abstract: "hormone use")
13	S	Med	PERC	Hormone use	exogenous oestrogen	Previous DVT/PE should say "objectively diagnosed" (label says "Previous DVT or PE")
24	S	Med	Wells	Item definitions	Stated in the parameter labels given to renderer and extractor: malignancy = on treatment, treated within 6 months, or palliative; DVT signs = leg swelling plus pain on deep-vein palpation; previous DVT/PE	
14	S	Med	Wells	Immobilisation duration	"> 3 days" (NICE)	"≥ 3 days" (MDCalc)
25	S	Med	Wells	4-week window	label: "Immobilisation > 3 days or surgery in the previous 4 weeks" (window read as applying to both)	the window may apply to surgery only
15	S	Med	Wells	PE-likely item	"alternative diagnosis less likely than PE" (NICE)	"PE #1 or equally likely" (MDCalc)
8	C	Med	CURB-65	BUN equivalent	code converts BUN to urea mmol/L and applies > 7 (= BUN > 19.6)	MDCalc: BUN > 19; an audit suggests > 20

#	Type	Priority	Calculator	Item	Implemented	Alternatives / notes
26	C	High	CURB-65 / qSOFA	How mental status is extracted (accepted: GCS 14 $\square$ present)	Bool params, labelled “Confusion (AMT $\leq$ 8 or new disorientation)” and “Altered mentation (GCS < 15)”. There is no GCS parameter: “GCS 14” $\square$ altered mentation present; a vague “drowsy” is left to the extractor’s judgement. Checked: dementia, baseline impairment and vague words (“lethargic”) appear only where mental status is explicitly documented, never where it is undocumented, so no case is ambiguous. Baseline impairment is consistent with each definition (qSOFA counts any abnormal mentation; CURB-65 counts only new disorientation)	add a numeric GCS parameter (would change prompts and cohort)
9	C	Low	CURB-65 / qSOFA	Confusion vs altered mentation	two separate parameters	Coupling is not needed: each case belongs to exactly one calculator, so one patient never carries both
10	C	Med	qSOFA	Altered mentation threshold	GCS < 15 (Sepsis-3)	GCS $\leq$ 13 (Seymour derivation model)
16	C	Med	Cockcroft-Gault	Weight	actual body weight (original)	ideal/adjusted weight in obesity; MedCalc-Bench picks by BMI (explains 49% anchor disagreement)
17	M	Low	Cockcroft-Gault	Decision thresholds	30, 60 mL/min (config-overridable)	FDA 2024 Table 1 also has 90
18	M	Low	Units	Creatinine factor	88.4 μ mol/L per mg/dL	88.42 (10,000 / 113.12). Clinically irrelevant; can move a boundary case by < 0.03% of CrCl. No case sits that close in our cohort
19	M	Med	Params	“Stated normal” intervals	see table below	—

#	Type	Priority	Calculator	Item	Implemented	Alternatives / notes
20	M	Low	Params	Creatinine can't be "normal"	Present or Unknown only	The no-straddle guarantee is per parameter and threshold. CG thresholds apply to a function of age, weight and creatinine, so no creatinine interval could guarantee a single CrCl band. For CG, bounds return a set of categories whenever an input is uncertain (100 of 200 CG cases are determined from the note by design)
21	M	Med	Cohort	Population priors	independent sampling (plus DBP < SBP – 15)	Independence can yield implausible combinations (e.g. established CAD with no risk factors, or age < 40 with ≥ 3 risk factors: 12 HEART cases, 6% of HEART and 1% of the cohort). It doesn't affect scores (atherosclerotic disease scores 2 regardless), and excluding the flagged cases leaves results unchanged (supplementary Table S10). Reported as a limitation; correlated priors are future work
22	C	Med	Cohort	Medication implying a comorbidity	Only metformin □ diabetes is reasonably specific. Lisinopril □ hypertension and atorvastatin □ hypercholesterolaemia are weak (ACE inhibitors in HF, post-MI and CKD; statins post-ACS and in diabetes). Scoring: for a comorbidity shown only through a medication, both Present and Unknown count as correct; only Absent is an error	treat the implication as probabilistic in the generator

S2. Population priors for the synthetic cohort

Plausible emergency-department populations for each calculator's intended use; parameters sampled independently except diastolic < systolic blood pressure – 15 mmHg. The PERC cohort represents low gestalt pre-test probability. Reviewed by the author.

Calculator	Parameter	Prior
HEART	heart_history	slightly_suspicious 0.45, moderately_suspicious 0.35, highly_suspicious 0.2
HEART	heart_ecg	normal 0.6, nonspecific_repolarization 0.3, significant_st_deviation 0.1
HEART	age	Normal(58, 15) truncated to [18, 100]
HEART	hypertension	P(yes) = 0.4
HEART	hypercholesterolemia	P(yes) = 0.3
HEART	diabetes	P(yes) = 0.2
HEART	obesity	P(yes) = 0.25
HEART	smoking	P(yes) = 0.25
HEART	family_history_cad	P(yes) = 0.2
HEART	atherosclerotic_disease	P(yes) = 0.2
HEART	heart_troponin	le_normal 0.75, 1_to_3x_normal 0.15, gt_3x_normal 0.1
CURB-65	confusion	P(yes) = 0.15
CURB-65	urea	Normal(7, 3.5) truncated to [1.5, 40]
CURB-65	resp_rate	Normal(22, 6) truncated to [8, 50]
CURB-65	sbp	Normal(125, 25) truncated to [60, 220]
CURB-65	dbp	Normal(72, 14) truncated to [30, 130]
CURB-65	age	Normal(65, 17) truncated to [18, 100]
qSOFA	resp_rate	Normal(21, 5) truncated to [8, 50]
qSOFA	altered_mentation	P(yes) = 0.2
qSOFA	sbp	Normal(118, 24) truncated to [60, 220]
PERC	age	Normal(38, 12) truncated to [18, 90]
PERC	heart_rate	Normal(88, 15) truncated to [45, 160]
PERC	spo2	Normal(97, 2) truncated to [85, 100]
PERC	unilateral_leg_swelling	P(yes) = 0.05
PERC	hemoptysis	P(yes) = 0.03
PERC	recent_surgery_trauma	P(yes) = 0.05
PERC	prior_vte	P(yes) = 0.05
PERC	hormone_use	P(yes) = 0.12
Wells PE	dvt_signs	P(yes) = 0.12
Wells PE	pe_most_likely	P(yes) = 0.3
Wells PE	heart_rate	Normal(95, 18) truncated to [45, 170]
Wells PE	immobilization_or_surgery	P(yes) = 0.12
Wells PE	prior_vte	P(yes) = 0.12
Wells PE	hemoptysis	P(yes) = 0.05
Wells PE	malignancy	P(yes) = 0.1
Cockcroft-Gault	age	Normal(60, 17) truncated to [18, 100]
Cockcroft-Gault	weight	Normal(75, 17) truncated to [35, 180]
Cockcroft-Gault	creatinine	Normal(1.3, 0.8) truncated to [0.4, 8]
Cockcroft-Gault	sex	male 0.5, female 0.5

S3. Missingness sensitivity (oracle extraction, natural share of undetermined cases)

Fresh cohorts (200 cases per calculator) at each missingness level; stratum results reweighted to the natural share of undetermined cases.

Missingness	System	Accuracy (natural share)	Questions/case (natural share)	Undetermined share
0.1	S1 Ask-all	0.999	0.642	0.252
0.1	S3-bin	0.964	0.089	0.252
	(missing=normal)			
0.1	S3 Bounds	0.999	0.326	0.252
0.1	S4 Bounds+VOI+echo	0.999	0.316	0.252
0.3	S1 Ask-all	0.997	1.925	0.61
0.3	S3-bin	0.899	0.282	0.61
	(missing=normal)			
0.3	S3 Bounds	0.997	1.144	0.61
0.3	S4 Bounds+VOI+echo	0.997	1.082	0.61
0.5	S1 Ask-all	0.994	3.289	0.824
0.5	S3-bin	0.824	0.483	0.824
	(missing=normal)			
0.5	S3 Bounds	0.994	2.159	0.824
0.5	S4 Bounds+VOI+echo	0.994	1.979	0.824

S4. Confirmation threshold sweep (S4 policy, clean notes, ideal clinician)

S4 asks the clinician to confirm a decision-critical extracted value when its calibrated confidence is below the threshold.

extraction	echo_threshold	accuracy	mean_questions	echo_per_case	echo_caught_errors
haiku__sonnet	0.5	0.995	0.8833	0.005	5
haiku__sonnet	0.8	0.9967	0.895	0.015	9
haiku__sonnet	0.9	0.9967	0.895	0.015	9
haiku__sonnet	0.95	0.9967	0.895	0.015	9
haiku__sonnet	0.99	0.9983	1.705	0.8242	19
haiku__sonnet	0.999	0.9983	1.7075	0.8267	19
qwen_local__sonnet	0.5	0.9983	1.1783	0.0	0
qwen_local__sonnet	0.8	0.9983	1.1783	0.0	0
qwen_local__sonnet	0.9	0.9983	1.1983	0.02	2
qwen_local__sonnet	0.95	0.9983	1.2967	0.1183	5
qwen_local__sonnet	0.99	0.9983	1.8142	0.6358	6
qwen_local__sonnet	0.999	0.9983	1.8142	0.6358	6

S5. Calibration of extraction confidence

Present/absent claims; isotonic and temperature calibration fitted on a seeded 30% development split, evaluated on the rest. Haiku 4.5 (self-reported): ECE 0.014 raw, 0.003 temperature, 0.002 isotonic. Qwen3.5-9B (token log-probabilities): ECE 0.048 raw, 0.033 temperature, 0.002 isotonic.

S6. Full results grid (559 paired cases; both extractors, both note styles, both clinicians)

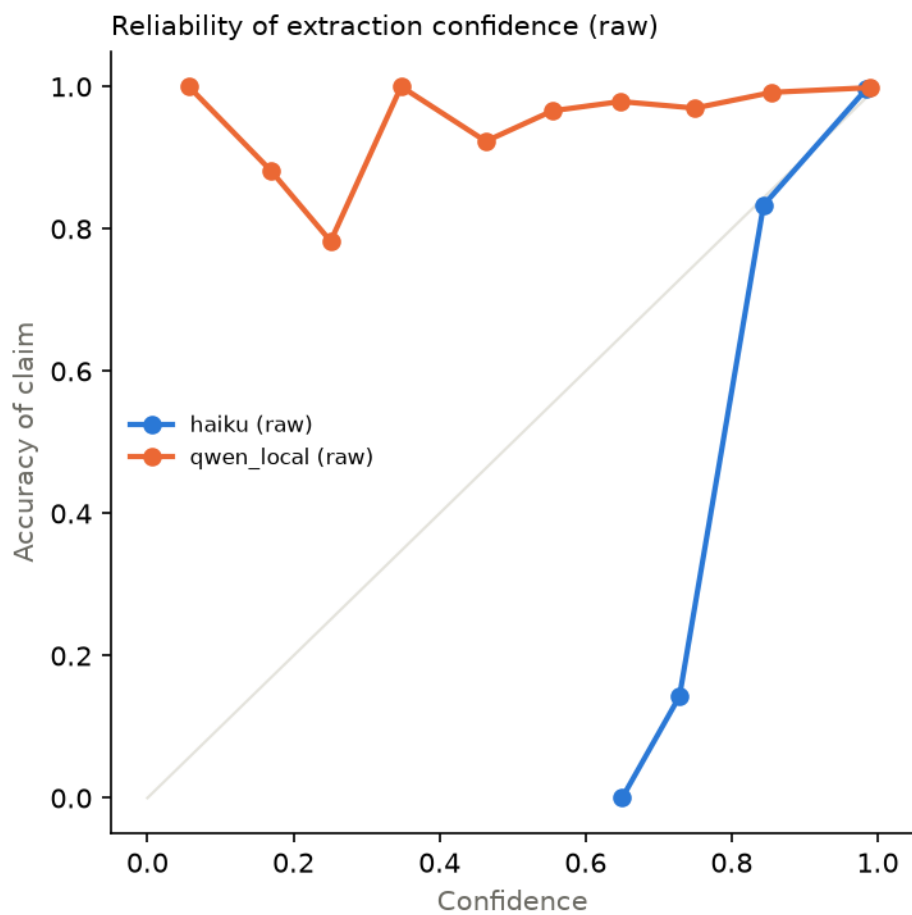


Figure 1: Reliability of raw extraction confidence (test split).

Condition	System	n	Accuracy, % (95% CI)	Correct when answering, %	Under- triage, % (95% CI)	Over- triage, %	Premature commit- ment, %	Questions/cases (95% CI)	Irrelevant responses, %
Haiku, clean notes, ideal clinician	S1 Ask-all	559	99.1 (97.9– 99.6)	99.3	0.2 (0.0–1.0)	0.7	0.0	1.71 (1.60– 1.81)	50.0
Haiku, clean notes, ideal clinician	S2 LLM agent	559	99.6 (98.7– 99.9)	99.8	0.2 (0.0–1.0)	0.2	0.2	0.92 (0.82– 1.01)	10.1
Haiku, clean notes, ideal clinician	S3 Bounds	559	99.1 (97.9– 99.6)	99.3	0.2 (0.0–1.0)	0.7	0.0	0.85 (0.75– 0.94)	0.0
Haiku, clean notes, ideal clinician	S4 Bounds+VOI+echocardiogram	559	99.6 (98.7– 99.9)	99.8	0.2 (0.0–1.0)	0.2	0.0	0.82 (0.73– 0.91)	0.0
Haiku, clean notes, ideal clinician	S3-bin (missing=normal)	559	91.4 (88.8– 93.5)	91.4	8.2 (6.2–10.8)	0.4	37.0	0.20 (0.16– 0.24)	0.0
Haiku, clean notes, noisy clinician	S1 Ask-all	559	88.2 (85.3– 90.6)	97.6	0.5 (0.2–1.6)	8.8	0.0	1.71 (1.60– 1.81)	48.6
Haiku, clean notes, noisy clinician	S2 LLM agent	559	84.4 (81.2– 87.2)	97.7	0.5 (0.2–1.6)	13.1	1.8	0.95 (0.84– 1.05)	8.5
Haiku, clean notes, noisy clinician	S3 Bounds	559	88.2 (85.3– 90.6)	97.6	0.5 (0.2–1.6)	8.8	0.0	0.88 (0.77– 0.97)	0.0
Haiku, clean notes, noisy clinician	S4 Bounds+VOI+echocardiogram	559	88.7 (85.8– 91.1)	98.0	0.5 (0.2–1.6)	8.2	0.0	0.86 (0.76– 0.96)	0.0
Haiku, clean notes, noisy clinician	S3-bin (missing=normal)	559	88.7 (85.8– 91.1)	91.2	8.2 (6.2–10.8)	2.5	36.0	0.20 (0.16– 0.25)	0.0
Haiku, messy notes, ideal clinician	S1 Ask-all	559	98.6 (97.2– 99.3)	98.7	0.5 (0.2–1.6)	0.9	0.0	1.92 (1.80– 2.04)	45.6
Haiku, messy notes, ideal clinician	S2 LLM agent	559	99.8 (99.0– 100.0)	100.0	0.0 (0.0–0.7)	0.2	0.7	0.92 (0.82– 1.01)	10.7
Haiku, messy notes, ideal clinician	S3 Bounds	559	98.6 (97.2– 99.3)	98.7	0.5 (0.2–1.6)	0.9	0.0	1.04 (0.94– 1.15)	0.0
Haiku, messy notes, ideal clinician	S4 Bounds+VOI+echocardiogram	559	99.1 (97.9– 99.6)	99.3	0.5 (0.2–1.6)	0.4	0.0	1.04 (0.94– 1.15)	0.0
Haiku, messy notes, ideal clinician	S3-bin (missing=normal)	559	90.7 (88.0– 92.8)	90.7	8.8 (6.7–11.4)	0.5	39.4	0.30 (0.23– 0.36)	0.0
Haiku, messy notes, noisy clinician	S1 Ask-all	559	85.5 (82.3– 88.2)	96.8	0.5 (0.2–1.6)	11.1	0.0	1.92 (1.80– 2.04)	43.6

Condition	System	n	Accuracy, % (95% CI)	Correct when answering, %	Under- triage, % (95% CI)	Over- triage, %	Premature commit- ment, %	Questions/cases (95% CI)	Irrelevant responses, %
Haiku, messy notes, noisy clinician	S2 LLM agent	559	85.0 (81.8– 87.7)	97.3	1.1 (0.5–2.3)	12.0	3.4	0.97 (0.87– 1.08)	10.1
Haiku, messy notes, noisy clinician	S3 Bounds	559	85.5 (82.3– 88.2)	96.8	0.5 (0.2–1.6)	11.1	0.0	1.08 (0.97– 1.20)	0.0
Haiku, messy notes, noisy clinician	S4 Bounds+VOI+echocardiography	559	85.9 (82.7– 88.5)	97.4	0.5 (0.2–1.6)	10.7	0.0	1.11 (1.00– 1.23)	0.0
Haiku, messy notes, noisy clinician	S3-bin (missing=normal)	559	86.8 (83.7– 89.3)	90.3	8.6 (6.5–11.2)	3.8	38.3	0.31 (0.25– 0.38)	0.0
Qwen, clean notes, ideal clinician	S1 Ask-all	559	99.8 (99.0– 100.0)	100.0	0.0 (0.0–0.7)	0.2	0.0	2.15 (2.04– 2.27)	45.5
Qwen, clean notes, ideal clinician	S2 LLM agent	559	99.6 (98.7– 99.9)	99.8	0.2 (0.0–1.0)	0.2	0.2	0.92 (0.82– 1.01)	10.1
Qwen, clean notes, ideal clinician	S3 Bounds	559	99.8 (99.0– 100.0)	100.0	0.0 (0.0–0.7)	0.2	0.0	1.17 (1.06– 1.28)	0.0
Qwen, clean notes, ideal clinician	S4 Bounds+VOI+echocardiography	559	99.8 (99.0– 100.0)	100.0	0.0 (0.0–0.7)	0.2	0.0	1.14 (1.04– 1.25)	0.0
Qwen, clean notes, ideal clinician	S3-bin (missing=normal)	559	91.6 (89.0– 93.6)	91.6	8.4 (6.4–11.0)	0.0	46.5	0.25 (0.21– 0.30)	0.0
Qwen, clean notes, noisy clinician	S1 Ask-all	559	84.4 (81.2– 87.2)	97.7	0.5 (0.2–1.6)	12.7	0.0	2.15 (2.04– 2.27)	44.0
Qwen, clean notes, noisy clinician	S2 LLM agent	559	84.4 (81.2– 87.2)	97.7	0.5 (0.2–1.6)	13.1	1.8	0.95 (0.84– 1.05)	8.5
Qwen, clean notes, noisy clinician	S3 Bounds	559	84.4 (81.2– 87.2)	97.7	0.5 (0.2–1.6)	12.7	0.0	1.21 (1.09– 1.32)	0.0
Qwen, clean notes, noisy clinician	S4 Bounds+VOI+echocardiography	559	84.4 (81.2– 87.2)	97.7	0.5 (0.2–1.6)	12.7	0.0	1.20 (1.09– 1.31)	0.0
Qwen, clean notes, noisy clinician	S3-bin (missing=normal)	559	87.8 (84.9– 90.3)	91.1	8.6 (6.5–11.2)	3.0	45.3	0.26 (0.22– 0.31)	0.0
Qwen, messy notes, ideal clinician	S1 Ask-all	559	99.5 (98.4– 99.8)	99.6	0.2 (0.0–1.0)	0.4	0.0	2.15 (2.03– 2.28)	45.7
Qwen, messy notes, ideal clinician	S2 LLM agent	559	99.8 (99.0– 100.0)	100.0	0.0 (0.0–0.7)	0.2	0.7	0.92 (0.82– 1.01)	10.7

Condition	System	n	Accuracy, % (95% CI)	Correct when answering, %	Under-triage, % (95% CI)	Over-triage, %	Premature commitment, %	Questions/cases, (95% CI)	Irrelevant questions, %
Qwen, messy notes, ideal clinician	S3 Bounds	559	99.5 (98.4–99.8)	99.6	0.2 (0.0–1.0)	0.4	0.0	1.17 (1.06–1.28)	0.0
Qwen, messy notes, ideal clinician	S4 Bounds+VOI+echo	559	99.6 (98.7–99.9)	99.8	0.0 (0.0–0.7)	0.4	0.0	1.22 (1.10–1.33)	0.0
Qwen, messy notes, ideal clinician	S3-bin (missing=normal)	559	90.0 (87.2–92.2)	90.0	9.8 (7.6–12.6)	0.2	47.6	0.23 (0.18–0.27)	0.0
Qwen, messy notes, noisy clinician	S1 Ask-all	559	84.3 (81.0–87.0)	97.5	0.4 (0.1–1.3)	12.7	0.0	2.15 (2.03–2.28)	43.7
Qwen, messy notes, noisy clinician	S2 LLM agent	559	85.0 (81.8–87.7)	97.3	1.1 (0.5–2.3)	12.0	3.4	0.97 (0.87–1.08)	10.1
Qwen, messy notes, noisy clinician	S3 Bounds	559	84.3 (81.0–87.0)	97.5	0.4 (0.1–1.3)	12.7	0.0	1.21 (1.09–1.33)	0.0
Qwen, messy notes, noisy clinician	S4 Bounds+VOI+echo	559	84.3 (81.0–87.0)	97.1	0.5 (0.2–1.6)	12.7	0.0	1.29 (1.17–1.41)	0.0
Qwen, messy notes, noisy clinician	S3-bin (missing=normal)	559	86.6 (83.5–89.2)	89.3	10.2 (8.0–13.0)	2.7	46.7	0.24 (0.19–0.28)	0.0

S7. Paired comparisons (exact McNemar for accuracy, Wilcoxon signed-rank for questions; Holm-adjusted within condition)

Full cohort (n = 1,200):

Condition	Comparison	Accuracy A vs B, %	Discordant (A only / B only)	McNemar p (Holm)	Questions A vs B	Wilcoxon p (Holm)
Oracle, clean, ideal	S3 Bounds vs S1 Ask-all	99.8 vs 99.8	0 / 0	1.000	0.92 vs 1.74	<0.001
Oracle, clean, ideal	S3 Bounds vs S2 LLM agent	99.8 vs 99.6	3 / 0	0.750	0.92 vs 0.99	<0.001
Oracle, clean, ideal	S3 Bounds vs S3-bin (missing=normal)	99.8 vs 91.8	99 / 2	<0.001	0.92 vs 0.21	<0.001
Oracle, clean, ideal	S4 Bounds+VOI+echo vs S3 Bounds	99.8 vs 99.8	0 / 0	1.000	0.87 vs 0.92	<0.001
Haiku, clean, ideal	S3 Bounds vs S1 Ask-all	99.4 vs 99.4	0 / 0	1.000	0.92 vs 1.78	<0.001
Haiku, clean, ideal	S3 Bounds vs S2 LLM agent	99.4 vs 99.6	3 / 5	1.000	0.92 vs 0.99	<0.001
Haiku, clean, ideal	S3 Bounds vs S3-bin (missing=normal)	99.4 vs 91.2	101 / 2	<0.001	0.92 vs 0.21	<0.001

Condition	Comparison	Accuracy A vs B, %	Discordant (A only / B only)	McNemar p (Holm)	Questions A vs B	Wilcoxon p (Holm)
Haiku, clean, ideal	S4 Bounds+VOI+echo vs S3 Bounds	99.7 vs 99.4	3 / 0	0.750	0.90 vs 0.92	0.005
Qwen, clean, ideal	S3 Bounds vs S1 Ask-all	99.8 vs 99.8	0 / 0	1.000	1.23 vs 2.21	<0.001
Qwen, clean, ideal	S3 Bounds vs S2 LLM agent	99.8 vs 99.6	3 / 0	0.750	1.23 vs 0.99	<0.001
Qwen, clean, ideal	S3 Bounds vs S3-bin (missing=normal)	99.8 vs 91.2	105 / 2	<0.001	1.23 vs 0.26	<0.001
Qwen, clean, ideal	S4 Bounds+VOI+echo vs S3 Bounds	99.8 vs 99.8	0 / 0	1.000	1.20 vs 1.23	0.004
Haiku, clean, noisy	S3 Bounds vs S1 Ask-all	87.0 vs 87.0	0 / 0	1.000	0.96 vs 1.78	<0.001
Haiku, clean, noisy	S3 Bounds vs S2 LLM agent	87.0 vs 83.5	65 / 23	<0.001	0.96 vs 1.03	<0.001
Haiku, clean, noisy	S3 Bounds vs S3-bin (missing=normal)	87.0 vs 88.0	70 / 82	0.750	0.96 vs 0.21	<0.001
Haiku, clean, noisy	S4 Bounds+VOI+echo vs S3 Bounds	87.2 vs 87.0	3 / 0	0.750	0.94 vs 0.96	0.078

Paired subset (n = 559):

Condition	Comparison	Accuracy A vs B, %	Discordant (A only / B only)	McNemar p (Holm)	Questions A vs B	Wilcoxon p (Holm)
Haiku, clean notes, ideal clinician	S3 Bounds vs S1 Ask-all	99.1 vs 99.1	0 / 0	1.000	0.85 vs 1.71	<0.001
Haiku, clean notes, ideal clinician	S3 Bounds vs S2 LLM agent	99.1 vs 99.6	1 / 4	0.750	0.85 vs 0.92	0.002
Haiku, clean notes, ideal clinician	S3 Bounds vs S3-bin (missing=normal)	99.1 vs 91.4	45 / 2	<0.001	0.85 vs 0.20	<0.001
Haiku, clean notes, ideal clinician	S4 Bounds+VOI+echo vs S3 Bounds	99.6 vs 99.1	3 / 0	0.750	0.82 vs 0.85	0.023
Haiku, clean notes, noisy clinician	S3 Bounds vs S1 Ask-all	88.2 vs 88.2	0 / 0	1.000	0.88 vs 1.71	<0.001
Haiku, clean notes, noisy clinician	S3 Bounds vs S2 LLM agent	88.2 vs 84.4	34 / 13	0.012	0.88 vs 0.95	<0.001
Haiku, clean notes, noisy clinician	S3 Bounds vs S3-bin (missing=normal)	88.2 vs 88.7	31 / 34	1.000	0.88 vs 0.20	<0.001
Haiku, clean notes, noisy clinician	S4 Bounds+VOI+echo vs S3 Bounds	88.7 vs 88.2	3 / 0	0.750	0.86 vs 0.88	0.342
Haiku, messy notes, ideal clinician	S3 Bounds vs S1 Ask-all	98.6 vs 98.6	0 / 0	1.000	1.04 vs 1.92	<0.001
Haiku, messy notes, ideal clinician	S3 Bounds vs S2 LLM agent	98.6 vs 99.8	0 / 7	0.047	1.04 vs 0.92	0.002
Haiku, messy notes, ideal clinician	S3 Bounds vs S3-bin (missing=normal)	98.6 vs 90.7	46 / 2	<0.001	1.04 vs 0.30	<0.001

Condition	Comparison	Accuracy A vs B, %	Discordant (A only / B only)	McNemar p (Holm)	Questions A vs B	Wilcoxon p (Holm)
Haiku, messy notes, ideal clinician	S4 Bounds+VOI+echo vs S3 Bounds	99.1 vs 98.6	3 / 0	0.500	1.04 vs 1.04	0.933
Haiku, messy notes, noisy clinician	S3 Bounds vs S1 Ask-all	85.5 vs 85.5	0 / 0	1.000	1.08 vs 1.92	<0.001
Haiku, messy notes, noisy clinician	S3 Bounds vs S2 LLM agent	85.5 vs 85.0	30 / 27	1.000	1.08 vs 0.97	0.032
Haiku, messy notes, noisy clinician	S3 Bounds vs S3-bin (missing=normal)	85.5 vs 86.8	33 / 40	1.000	1.08 vs 0.31	<0.001
Haiku, messy notes, noisy clinician	S4 Bounds+VOI+echo vs S3 Bounds	85.9 vs 85.5	3 / 1	1.000	1.11 vs 1.08	0.048
Qwen, clean notes, ideal clinician	S3 Bounds vs S1 Ask-all	99.8 vs 99.8	0 / 0	1.000	1.17 vs 2.15	<0.001
Qwen, clean notes, ideal clinician	S3 Bounds vs S2 LLM agent	99.8 vs 99.6	1 / 0	1.000	1.17 vs 0.92	<0.001
Qwen, clean notes, ideal clinician	S3 Bounds vs S3-bin (missing=normal)	99.8 vs 91.6	47 / 1	<0.001	1.17 vs 0.25	<0.001
Qwen, clean notes, ideal clinician	S4 Bounds+VOI+echo vs S3 Bounds	99.8 vs 99.8	0 / 0	1.000	1.14 vs 1.17	0.033
Qwen, clean notes, noisy clinician	S3 Bounds vs S1 Ask-all	84.4 vs 84.4	0 / 0	1.000	1.21 vs 2.15	<0.001
Qwen, clean notes, noisy clinician	S3 Bounds vs S2 LLM agent	84.4 vs 84.4	32 / 32	1.000	1.21 vs 0.95	<0.001
Qwen, clean notes, noisy clinician	S3 Bounds vs S3-bin (missing=normal)	84.4 vs 87.8	34 / 53	0.212	1.21 vs 0.26	<0.001
Qwen, clean notes, noisy clinician	S4 Bounds+VOI+echo vs S3 Bounds	84.4 vs 84.4	0 / 0	1.000	1.20 vs 1.21	0.960
Qwen, messy notes, ideal clinician	S3 Bounds vs S1 Ask-all	99.5 vs 99.5	0 / 0	1.000	1.17 vs 2.15	<0.001
Qwen, messy notes, ideal clinician	S3 Bounds vs S2 LLM agent	99.5 vs 99.8	0 / 2	1.000	1.17 vs 0.92	<0.001
Qwen, messy notes, ideal clinician	S3 Bounds vs S3-bin (missing=normal)	99.5 vs 90.0	54 / 1	<0.001	1.17 vs 0.23	<0.001
Qwen, messy notes, ideal clinician	S4 Bounds+VOI+echo vs S3 Bounds	99.6 vs 99.5	1 / 0	1.000	1.22 vs 1.17	0.006
Qwen, messy notes, noisy clinician	S3 Bounds vs S1 Ask-all	84.3 vs 84.3	0 / 0	1.000	1.21 vs 2.15	<0.001
Qwen, messy notes, noisy clinician	S3 Bounds vs S2 LLM agent	84.3 vs 85.0	31 / 35	1.000	1.21 vs 0.97	<0.001
Qwen, messy notes, noisy clinician	S3 Bounds vs S3-bin (missing=normal)	84.3 vs 86.6	42 / 55	0.892	1.21 vs 0.24	<0.001
Qwen, messy notes, noisy clinician	S4 Bounds+VOI+echo vs S3 Bounds	84.3 vs 84.3	2 / 2	1.000	1.29 vs 1.21	<0.001

S8. Real case reports, all splits and extractors

Split	Extractor	Calculator	n	Code reproduces label, %	Determined from note, % (95% CI)	Truth still possible, %	Category correct, missing=normal, %	Entity agreement, %
train	Haiku 4.5	Cockcroft-Gault	125	51	88.0 (81.1–92.6)	85	71	90
train	Haiku 4.5	CURB-65	85	100	49.4 (39.0–59.8)	93	82	89
train	Haiku 4.5	HEART	125	100	12.8 (8.0–19.8)	90	22	73
train	Haiku 4.5	PERC	124	100	90.3 (83.8–94.4)	100	94	90
train	Haiku 4.5	Wells PE	125	100	19.2 (13.3–27.0)	98	81	87
train	Haiku 4.5	All	584	90	52.1 (48.0–56.1)	93	69	86
train	Qwen3.5-9B	Cockcroft-Gault	125	51	84.8 (77.5–90.0)	90	72	92
train	Qwen3.5-9B	CURB-65	85	100	41.2 (31.3–51.8)	95	81	92
train	Qwen3.5-9B	HEART	125	100	12.0 (7.4–18.9)	93	9	74
train	Qwen3.5-9B	PERC	124	100	87.9 (81.0–92.5)	100	91	89
train	Qwen3.5-9B	Wells PE	125	100	9.6 (5.6–16.0)	99	78	88
train	Qwen3.5-9B	All	584	90	47.4 (43.4–51.5)	95	65	87
test	Haiku 4.5	Cockcroft-Gault	20	40	95.0 (76.4–99.1)	100	95	96
test	Haiku 4.5	CURB-65	20	100	55.0 (34.2–74.2)	100	100	97
test	Haiku 4.5	HEART	20	95	60.0 (38.7–78.1)	85	45	87
test	Haiku 4.5	PERC	20	100	75.0 (53.1–88.8)	100	95	99
test	Haiku 4.5	Wells PE	20	100	10.0 (2.8–30.1)	100	85	89
test	Haiku 4.5	All	100	87	59.0 (49.2–68.1)	97	84	94
test	Qwen3.5-9B	Cockcroft-Gault	20	40	90.0 (69.9–97.2)	100	90	96
test	Qwen3.5-9B	CURB-65	20	100	50.0 (29.9–70.1)	95	90	98
test	Qwen3.5-9B	HEART	20	95	30.0 (14.5–51.9)	95	35	77
test	Qwen3.5-9B	PERC	20	100	75.0 (53.1–88.8)	100	95	96
test	Qwen3.5-9B	Wells PE	20	100	10.0 (2.8–30.1)	100	85	89

Split	ExtractoCalculator	n	Code reproduces label, %	Determined from note, % (95% CI)	Truth still possible, %	Category correct, missing=normal, %	Entity agreement, %
test	Qwen3.5-72B	100	87	51.0 (41.3–60.6)	98	79	91

S10. Sensitivity to implausible synthetic cases

Characteristics were sampled independently, so some cases are clinically implausible. We flagged established atherosclerotic disease with no risk factors, and age under 40 with three or more risk factors or atherosclerotic disease. We then recomputed the main results without those cases (clean notes, ideal clinician).

12 of 1200 cases flagged: heart-0017 (age < 40 with ≥ 3 risk factors or atherosclerotic disease); heart-0023 (age < 40 with ≥ 3 risk factors or atherosclerotic disease); heart-0036 (age < 40 with ≥ 3 risk factors or atherosclerotic disease); heart-0069 (age < 40 with ≥ 3 risk factors or atherosclerotic disease); heart-0087 (established atherosclerotic disease with no risk factors); heart-0104 (established atherosclerotic disease with no risk factors); heart-0128 (established atherosclerotic disease with no risk factors); heart-0132 (established atherosclerotic disease with no risk factors); heart-0135 (age < 40 with ≥ 3 risk factors or atherosclerotic disease); heart-0137 (established atherosclerotic disease with no risk factors); heart-0172 (age < 40 with ≥ 3 risk factors or atherosclerotic disease); heart-0195 (established atherosclerotic disease with no risk factors)

Extraction	Cases	System	n	Accuracy, %	Under-triage, %	Questions/case
Oracle	all	S1 Ask-all	1200	99.8	0.0	1.74
Oracle	all	S2 LLM agent	1200	99.6	0.1	0.99
Oracle	all	S3 Bounds	1200	99.8	0.0	0.92
Oracle	all	S4 Bounds+VOI+echo	1200	99.8	0.0	0.87
Oracle	all	S3-bin (missing=normal)	1200	91.8	8.2	0.21
Oracle	excluding flagged	S1 Ask-all	1188	99.8	0.0	1.73
Oracle	excluding flagged	S2 LLM agent	1188	99.6	0.1	0.99
Oracle	excluding flagged	S3 Bounds	1188	99.8	0.0	0.92
Oracle	excluding flagged	S4 Bounds+VOI+echo	1188	99.8	0.0	0.87
Oracle	excluding flagged	S3-bin (missing=normal)	1188	91.9	8.1	0.21
Haiku 4.5	all	S1 Ask-all	1200	99.4	0.1	1.78
Haiku 4.5	all	S2 LLM agent	1200	99.6	0.1	0.99
Haiku 4.5	all	S3 Bounds	1200	99.4	0.1	0.92
Haiku 4.5	all	S4 Bounds+VOI+echo	1200	99.7	0.1	0.90
Haiku 4.5	all	S3-bin (missing=normal)	1200	91.2	8.5	0.21
Haiku 4.5	excluding flagged	S1 Ask-all	1188	99.4	0.1	1.77
Haiku 4.5	excluding flagged	S2 LLM agent	1188	99.6	0.1	0.99
Haiku 4.5	excluding flagged	S3 Bounds	1188	99.4	0.1	0.93
Haiku 4.5	excluding flagged	S4 Bounds+VOI+echo	1188	99.7	0.1	0.90
Haiku 4.5	excluding flagged	S3-bin (missing=normal)	1188	91.3	8.3	0.21
Qwen3.5-72B	all	S1 Ask-all	1200	99.8	0.0	2.21
Qwen3.5-72B	all	S2 LLM agent	1200	99.6	0.1	0.99

Extraction	Cases	System	n	Accuracy, %	Under-triage, %	Questions/case
Qwen3.5-9B	all	S3 Bounds	1200	99.8	0.0	1.23
Qwen3.5-9B	all	S4 Bounds+VOI+echo	1200	99.8	0.0	1.20
Qwen3.5-9B	all	S3-bin (missing=normal)	1200	91.2	8.8	0.26
Qwen3.5-9B	excluding flagged	S1 Ask-all	1188	99.8	0.0	2.20
Qwen3.5-9B	excluding flagged	S2 LLM agent	1188	99.6	0.1	0.99
Qwen3.5-9B	excluding flagged	S3 Bounds	1188	99.8	0.0	1.23
Qwen3.5-9B	excluding flagged	S4 Bounds+VOI+echo	1188	99.8	0.0	1.20
Qwen3.5-9B	excluding flagged	S3-bin (missing=normal)	1188	91.4	8.6	0.26

S9. Exact prompts

Prompt versions: renderer v3, extractor x1, agent a1. The example fact sheet is for one CURB-65 case with a trap; <the rendered note> stands for the note text.

Renderer (system)

You write realistic clinical notes for a research dataset. Every patient is fictional; the notes are

Renderer (user), example CURB-65 case

Write one clinical note in US English (en-US).

Setting: Emergency department. Adult with suspected community-acquired pneumonia.

Style guide:

- US emergency department or inpatient documentation conventions.
- Typical sections: Chief complaint, HPI, PMH, Medications, Allergies, Social history, Exam (vitals first), Results, Assessment and plan. Not every section is required.
- Common US abbreviations are fine (HPI, PMH, SOB, CP, ROS, HR, RR, BP, SpO2, NKDA).
- Units: US conventional (mg/dL, lb or kg, °F or °C) except where the fact sheet gives a unit.
- Terse clinical style; sentence fragments are normal.

Fact sheet (each item is mandatory):

- [confusion] STATE: the patient is alert and fully oriented, not confused.
- [urea] STATE (number exactly): serum urea 8 mmol/L. Also mention a Serum urea (mmol/L) of 6.6 fr
- [resp_rate] DO NOT MENTION Respiratory rate (/min). Include nothing from which it could be infer
- [sbp] DO NOT MENTION Systolic blood pressure (mmHg). Include nothing from which it could be infe
- [dbp] DO NOT MENTION Diastolic blood pressure (mmHg). Include nothing from which it could be inf
- [age] DO NOT MENTION Age (years). Include nothing from which it could be inferred (no value, no

Rules:

- Write numbers exactly as given in the fact sheet (same digits and decimals), with the unit given
- Express each item in natural clinical language, as a clinician would chart it; do not copy the f
- Every STATE item must appear explicitly in the note, including negative ones (e.g. "no diabetes"
- Follow each item exactly. Do not add information about any fact-sheet item beyond what it says,
- No blanket or complete statements that would cover a DO NOT MENTION item: no "Medications: none"
- You may add other realistic details (presenting complaint, unrelated history, unrelated examinatio
- Do not name any clinical score or rule (HEART score, CURB-65, CURB65, qSOFA, PERC, Wells, Cockcro
- Do not refer to these instructions, the fact sheet, or the note being synthetic.
- Output only the note, 150-350 words, plain text (simple section headings allowed).

Judge (system)

You audit synthetic clinical notes against a list of parameters. For each parameter, decide only f

Judge (user)

Note:

<note>

<the rendered note>

</note>

Parameters:

- confusion: Confusion (AMT $\leq$ 8 or new disorientation to person/place/time) [yes/no finding]
- urea: Serum urea (mmol/L) [numeric (mmol/L)]
- resp_rate: Respiratory rate (/min) [numeric (/min)]
- sbp: Systolic blood pressure (mmHg) [numeric (mmHg)]
- dbp: Diastolic blood pressure (mmHg) [numeric (mmHg)]
- age: Age (years) [numeric (years)]

For every parameter, return one object:

{"param": "<id>", "status": "stated" | "negated_or_normal" | "not_mentioned" | "implied", "level":

- stated: the value or finding is explicitly documented (for yes/no findings: present).
- negated_or_normal: explicitly absent, denied, or documented as normal without a value.
- not_mentioned: nothing in the note documents or implies it.
- implied: not explicitly documented, but inferable from the note.

Return a JSON array with exactly one object per parameter, in the order listed, and nothing else.

Extractor (system)

You extract clinical parameters from a clinical note into a fixed JSON structure. You report only

Extractor (user)

Note:

<note>

<the rendered note>

</note>

Parameters:

- confusion: Confusion (AMT $\leq$ 8 or new disorientation to person/place/time) [yes/no finding]
- urea: Serum urea (mmol/L) [numeric; unit tokens: mmol/L, urea_mg/dL, bun_mg/dL]

- resp_rate: Respiratory rate (/min) [numeric; unit tokens: /min]
- sbp: Systolic blood pressure (mmHg) [numeric; unit tokens: mmHg]
- dbp: Diastolic blood pressure (mmHg) [numeric; unit tokens: mmHg]
- age: Age (years) [numeric; unit tokens: years]

For every parameter give:

- "status":
 - "present": the note states the finding, or states a value/level for it.
 - "absent": the note explicitly says the finding is absent or denied, or that the parameter is not applicable.
 - "unknown": the note does not document it. Do not infer absence from silence: if the note does not mention the parameter, it is unknown.
- "value": for yes/no findings use null; for graded parameters one of the listed levels; for numeric parameters the value as written, mapped to one of the listed unit tokens; otherwise null.
- "unit": for numeric parameters, the unit as written, mapped to one of the listed unit tokens; otherwise null.
- "evidence": the shortest exact quote from the note (copied character for character) that supports the status and value.
- "confidence": your probability (0 to 1) that status and value are correct.

If the note gives several values for the same parameter (e.g. an earlier reading or a previous visit), use the most recent one.

S2 agent (system)

You are a clinical decision-support agent working with an emergency clinician. You determine a patient's risk of sepsis based on the CURB-65 calculator.

S2 agent (first user turn)

Calculator: CURB-65

Decision categories: low, moderate, high

Clinical note:

<note>

<the rendered note>

</note>

Calculator parameters (ids you can ask about or pass to calculate):

- confusion: Confusion (AMT $\leq$ 8 or new disorientation to person/place/time) [yes/no]
- urea: Serum urea (mmol/L) [number in mmol/L]
- resp_rate: Respiratory rate (/min) [number in /min]
- sbp: Systolic blood pressure (mmHg) [number in mmHg]
- dbp: Diastolic blood pressure (mmHg) [number in mmHg]
- age: Age (years) [number in years]

Actions (reply with one JSON object per turn):

1. {"action": "ask", "parameter": "<id>", "question": "<your question to the clinician>"}
The clinician answers from the patient (or says the value is not available).
2. {"action": "calculate", "values": {"<id>": <value>, ...}}
Runs the calculator. Give every parameter: yes/no as true/false, graded ones as the level name, and numeric ones as the value.
3. {"action": "answer", "category": "<one of: low, moderate, high, cannot_determine>"}
Your final decision category. Use "cannot_determine" only if it truly cannot be determined.